Rhea Senthil Kumar

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Education

University of California San Diego Expected December 2026
M.S. Computer Science and Engineering, Artificial Intelligence | Overall GPA: 4.0

University of California, Berkeley May 2023
B.A Computer Science & Astrophysics | Overall GPA: 3.8

Highlighted Research and Work Experience

UCSD, Gravitational Wave Paleontology Group December 2025 - Present
Graduate Student Researcher, Advisor: Dr. Floor Broekgaarden

- **Generative AI for Physics Simulation:** Developed novel generative AI models to emulate computationally intensive gravitational-wave population synthesis pipelines.
- **Conditional Flow Matching (CFM):** Introduced the first application of CFM for gravitational-wave population emulation, outperforming diffusion-based models and standard probabilistic baselines in capturing complex multi-modal distributions.
- **Scaling & Efficiency:** Achieved **300–500**× acceleration in population generation while maintaining strict physical fidelity (KL divergence **~0.01**) on simulated test distributions.

Qualcomm, SMTR Device Integration June 2026 - September 2026
Engineering Intern, Supervisors: Ciara Martin, Alan Shinsato

- **Test Hardware Integration:** Developed from scratch a Python library integrating Signal Hound signal generators and spectrum analyzers into the QGov modem team's testing framework, adopted team-wide as a drop-in replacement with an unchanged user workflow.
- **Cost Reduction:** Enabled test benches built on 250× cheaper equipment while keeping full legacy hardware support.
- **Validation & Parity:** Kept test performance within factory bounds, achieving identical pass/fail results across the new and legacy test beds.

MIT Lincoln Laboratory, Space Systems Analysis and Test Group September 2023 - September 2025
Assistant Staff, Supervisor: Eva Nickelson

- **Systems Analysis & Red Teaming:** Conducted systems-level analysis and adversarial evaluation of space-domain systems for national security. Maintained active **TS//SCI clearance**, read out September 2025.
- **Generative Data Augmentation:** Built deep generative models to synthesize high-fidelity radar imagery, mitigating severe class imbalances and data sparsity for downstream classification networks.
- **Performance Engineering:** Engineered a distributed signal cross-correlation pipeline for grid processing systems, boosting high-throughput data ingestion by **80**×.

UC Berkeley Space Sciences Laboratory, GAPS Integration and Testing August 2022 - August 2023
Research Apprentice, Advisor: Dr. Field Rogers

- Campaigned to conduct integrated thermal vacuum tests on General AntiParticle Spectrometer (GAPS).
- Developed software tools for GAPS integration and calibration tests in preparation for 2024 deployment.

UC Berkeley Space Sciences Laboratory, COSI Collaboration January 2020 - August 2023
Research Assistant and Lab Lead, Advisors: Dr. Andreas Zoglauer, Dr. John Tomsick

- **Deep Learning Reconstruction:** Built custom deep learning architectures in TensorFlow/Keras for spatial event reconstruction, driving a **27%** improvement in classification accuracy to achieve a **99%** baseline.

- **Open-Source Infrastructure:** Authored an end-to-end simulation, calibration, and ML inference pipeline adopted globally by NASA and UC Berkeley high-energy astrophysics teams.

NASA Goddard Space Flight Center, AMEGO-X Simulation Analysis June 2021 - August 2021
NASA Intern, Advisors: Dr. Carolyn Kierans, Dr. Henrike Fleischhack

- Developed Kalman filter pair reconstruction algorithm and validated performance for All-Sky Medium-Energy Gamma Ray eXplorer.
- Fitted spatial distribution of energy in AMEGO-X calorimeter to determine energy resolution in pair regime.

Selected Projects

From-Scratch 100M-Parameter Language Model | PyTorch, Distributed Training May 2026

- Implemented a nanoGPT-style decoder-only Transformer from scratch, optimized strictly within a 100M-parameter threshold on the FineWeb-Edu corpus. Achieved **25.8** perplexity on the held out test set.
- Integrated modern architectural features: Grouped Query Attention (GQA), Rotary Position Embeddings (RoPE), SwiGLU activations, QK normalization, and weight tying.
- Engineered an end-to-end training pipeline using GPT-2 BPE tokenizer; implemented progressive context-length curriculum and knowledge-distillation training, and tracked loss convergence dynamics.

Generative Audio & Automatic Music Transcription System | PyTorch, Librosa May 2026

- Developed an end-to-end neural transcription pipeline mapping raw acoustic waveforms to symbolic MIDI data by combining multi-scale spectrogram backbones with Transformer sequence models.
- Built a Transformer-based unconditioned symbolic music generation model that outperformed CNN and LSTM baselines in sequence coherence and musical structure preservation on the MAESTRO dataset.
- Developed a Conformer-based automatic music transcription architecture that outperformed CRNN and CQT baseline models across key transcription metrics, achieving best frame/note-level F1 scores, ghost note and missed note rates through joint convolutional and self-attention feature learning.

Memory Architectures for Long-Horizon LLM Agents | Agent Evaluation, Memory Systems March 2026

- Developed a benchmark and evaluation framework for understanding how different memory mechanisms shape long-horizon LLM agent capabilities. Submitted to NeurIPS 2026.
- Implemented semantic, working, procedural, and episodic memory architectures, including graph-based world models, ReAct reasoning agents, VQ-VAE latent action models, and compressed memory retrieval systems.
- Proposed hybrid memory agents that integrate complementary cognitive capabilities and evaluated trade-offs between specialization, robustness, and computational efficiency.
- Characterized failure modes in autonomous agents, including memory degradation, repetitive action loops, retrieval errors, and limitations of single-memory architectures.

Fair Resource Allocation in *Settlers of Catan* | Python, CVXPY, Multi-Agent Systems March 2026

- Conceived and led the project, formulating a convex optimization framework for equitable resource allocation in *Settlers of Catan* through continuous relaxation, opponent-aware fairness constraints, and phase-adaptive production objectives.
- Designed and implemented a quadratic-programming (QP) agent from scratch, deriving the optimization formulation, constraints, and KKT conditions while integrating the policy into a custom multi-agent simulation framework built on the Catanatron engine.
- Engineered a reproducible evaluation pipeline spanning **35 agent lineups** and **17,500 simulated games**, demonstrating reduced outcome inequality (lower Victory Point Gini coefficient) while characterizing trade-offs between fairness, resource diversity, computational cost, and competitive performance.

Anime Recommendation System | PyTorch, Bayesian Personalized Ranking (BPR), Python December 2025

- Designed a hybrid recommendation engine for personalized top-K anime retrieval by combining implicit Bayesian Personalized Ranking (BPR), collaborative filtering, content-based metadata embeddings, and popularity-based cold-start fallbacks.
- Engineered a scalable data preprocessing pipeline for **109M+** user–anime interactions across **310K** users and **17K** titles, including feature engineering, negative sampling, metadata encoding, and train/validation/test dataset construction.
- Evaluated ranking performance using Precision@K, Recall@K, Hit Rate@K, and AUC, achieving **0.95** validation AUC while demonstrating improved personalized recommendations over popularity-based and item-based collaborative filtering baselines.

GNN Sensor Tasking for Space Domain Awareness | PyTorch, PyTorch Geometrics, PPO, RL June 2025

- Developed a graph neural network reinforcement learning framework for autonomous sensor tasking, modeling ground sensors and orbital targets as graph nodes to optimize multi-sensor coordination in dynamic tracking environments.
- Designed a PPO-based policy with graph attention networks and recurrent memory to balance search and track objectives under limited sensor resources, enabling adaptive decision-making across varying target densities.
- Built a configurable orbital simulation environment and benchmarked the learned policy against heuristic scheduling baselines, achieving **99%** target tracking coverage and consistently meeting the <10-minute mission search constraint for maneuvering targets (**38%** improvement over heuristic baselines), while all non-learning baselines failed to satisfy this real-time performance requirement in oversaturated scenarios.

Computer Graphics Physically-Based Renderer | C++ May 2023

- Implemented a complete graphics pipeline from scratch, including triangle rasterization, supersampling anti-aliasing, perspective-correct texture mapping, mipmapping, and hierarchical transformations.
- Developed physically based Monte Carlo path tracer featuring ray-scene intersection, BVH acceleration structures, global illumination, importance sampling, microfacet BRDFs, environment lighting, and adaptive sampling.
- Built geometry processing and physics simulation systems, implementing half-edge mesh data structures, Bezier curves and surfaces, Loop subdivision, mesh editing operations, and real-time mass-spring cloth simulation with collision handling and GLSL shading.

Publications and Presentations

NeurIPS AI4Science Workshop, Sydney, Australia, December 2026. R.S. Kumar, F. Broekgaarden. “PLANT: Conditional Generative Models for Fast Gravitational-Wave Population Synthesis” (accepted poster).

NeurIPS Women in ML Workshop, Atlanta, GA, December 2026. R.S. Kumar, F. Broekgaarden. “PLANT: Conditional Generative Models for Fast Gravitational-Wave Population Synthesis” (accepted poster).

GalFRESCA, Irvine, CA, September 2026. R. S. Kumar, F. Broekgaarden. “PLANT: Conditional Generative Models for Fast Gravitational-Wave Population Synthesis” (talk).

Ensenada-San Diego Astrophysical Society (ESDAS) Meeting, Ensenada, Mexico, September 2026. R. S. Kumar, F. Broekgaarden. “PLANT: Conditional Generative Models for Fast Gravitational-Wave Population Synthesis” (accepted talk).

Fast Machine Learning for Science Conference, San Diego, CA, August 2026. R. S. Kumar, F. Broekgaarden. “PLANT: Conditional Generative Models for Fast Gravitational-Wave Population Synthesis” (poster).

IAIFI Summer Workshop, Cambridge, MA, August 2026. R. S. Kumar, F. Broekgaarden. “From Thousands of CPU-Hours to Seconds: Fast Generative Models for Gravitational-Wave Populations” (talk).

AstroAI Workshop, Cambridge, MA, June 2026. R. S. Kumar, F. Broekgaarden. “PLANT: Conditional Generative Models for Fast Gravitational-Wave Population Synthesis” (poster).

IEEE High Performance Extreme Computing Conference, Virtual, September 2024. R.S. Kumar et. al. “Impact of Grid Processing on Signal Cross-Correlation” (poster).

American Astronomical Society, 241st Meeting, Seattle, WA, January 2023. H. Gulick et. al. “The Background and Transient Observer: A Gamma-Ray Detector System for Time-Domain Astrophysics” (presentation).

IEEE NSS MIC RTSD Conference, Milan, Italy, November 2022. R. S. Kumar, Y. Ebrahimi, O. Salaben, A. Zoglauer. “Evaluation of Compton Event Reconstruction Approaches using Simulations, Lab Measurements, and Astrophysical Observations of COSI” (talk).

Baylearn - Machine Learning Symposium, San Francisco, CA, October 2022. Y. Ebrahimi, O. Salaben, R. S. Kumar, A. Zoglauer. “Physics-based Validation of Machine-learning Approaches for the COSI Space Mission” (poster).

American Astronomical Society, 237th Meeting, Virtual, January 2021. A. Zoglauer et. al. “Enhancing the Event Reconstruction Pipeline of Future Compton-scattering and Pair-creation Telescopes with Deep Learning” (poster).

Honors

- AIAA New England Section Leadership Award — June 2025
- MIT Advanced Study Program Fellow — September 2024
- Berkeley Undergraduate Research Apprenticeship Program Summer Fellow — April 2023
- National Merit Scholarship — May 2019
- Dean's List - College Of Letters & Science, University of California, Berkeley

Leadership/Service

MIT BeaverWorks Summer Institute, Forge AI Challenge — March 2026 - Present
Instructor and Program Lead

- Founded a national online educational program teaching high school students modern Generative AI engineering, scaling, and safety principles.
- Designed a comprehensive interactive curriculum from scratch covering Transformer mechanics, foundational LLM training loops, Retrieval-Augmented Generation (RAG), multi-agent orchestrations, and AI safety/alignment paradigms.
- Directing technical course deployment, project infrastructure, and managing collaborations across MIT faculty and industry researchers.

MIT BeaverWorks Summer Institute, Build a CubeSat Challenge — February 2024 - Present
Instructor and Program Lead

- Lead a national engineering program mentoring student teams through full-lifecycle spacecraft development and edge-computing payloads.
- Designed software curriculum focusing on flight software deployment, embedded environments, and low-latency real-time computer vision pipelines hosted on resource-constrained Raspberry Pi hardware.
- Coordinate a nationwide team of instructional staff and industry collaborators spanning NASA, MIT Lincoln Laboratory, and aerospace research groups.

UC Berkeley Democratic Education at Cal (DeCal) Program September 2022 - May 2023
DeCal Creator and Facilitator

- Instructor of student-led course held during the Spring 2023 semester at UC Berkeley. Educated students on the sociopolitical context and art history of Studio Ghibli films and included showings.

UC Berkeley New Student Services (NSS) March 2020 - August 2022
Orientation Mentor and Leader

- Oversaw training of orientation group leaders. Trained in leadership methods, conflict management, collaborative methods, and networking.
- Managed events at orientation representing NSS. Facilitated discussion around diversity, inclusion, safety, and more at UC Berkeley.

Membership

- MIT Lincoln Laboratory Pan Asian Laboratory Staff Network (2023-2025)
- Undergraduate Astronomy Society Berkeley (2019-2023)
- UC Berkeley Society of Physics Students (2019-2023)

Relevant Coursework:
UCSD: Machine Learning for Music, Recommender Systems & Web Mining, Convex Optimization, AI Agents, Probabilistic Reasoning & Learning, Deep Learning
UC Berkeley: Artificial Intelligence, Intro to Machine Learning, Computer Graphics, Linear Algebra & Differential Equations, Multivariable Calculus, Discrete Math & Probability, Data Structures, Intro to CS Theory

Technical Skills

- **Languages:** Python, C++, C, Java, Javascript, Go, SQL, HTML, MATLAB, Scheme
- **Machine Learning:** PyTorch, TensorFlow, Keras, Neural Network Training and Architecture Design, Conditional Flow Matching (CFM), Diffusion Models, Large Language Models (LLMs), Transformers (GQA/RoPE), Reinforcement Learning (PPO), Graph Neural Networks (GNNs), Computer Vision, Probabilistic Modeling, Model Context Protocol (MCP), Agentic Guardrails, Multimodal AI Systems
- **Tools & Systems:** Distributed Computing, Git, Linux, Raspberry Pi, Shell Scripting, PyQt, Geant4, Data Visualization, GUI Development, TMVA, Latex, ROOT, AutoCAD, Technical Briefing, GUI Development
- **Laboratory:** Systems Integration, Instrumentation/Payload Testing and Calibration, Mechanical Assembly, Pipeline Development, Optical Telescope Operation, Infrared Telescope Operation, Radiation Material Handling, Thermal/Vacuum Chamber Usage, Instrument Commissioning Operations, Analog and Digital Electronics, Circuit Fabrication and Testing